

Alex Rother

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Computer Science undergraduate interested in application development, cybersecurity, computer networking, and web development.

EDUCATION

Bachelor of Science in Computer Science

University of Tennessee, Knoxville, Tennessee

Fall 2023 – Spring 2027

3.0

EXPERIENCE

Student Worker, OIT Networking

May 2026 – Present

University of Tennessee, Knoxville, Tennessee

- **Arista switch** installation, patching, and config.
- Wrote software for **Arista EOS** to further automate switch configuration.

Student Intern, IT Assistant

Fall 2020 – Spring 2021

Le Jardin Academy, Kailua, Hawaii

- Chromebook **hardware repairs** and classroom device troubleshooting.
- General IT tasks, software installation, and troubleshooting managed devices.

PROJECTS

Application Development

Hypixel API, Discord API, Java Swing, Unity

- Wrote a web based **Remote Monitoring and Management** panel and cross-platform Rust agent.
- Developed a **multithreaded** Windows **auto clicker** using C++ and the Windows API.
- Developed a **graphical calculator** using Java Swing.
- Developed a **2D game** with procedural dungeon generation in Unity.

Cybersecurity

Win API, Donut, Promon, Binary Ninja, Nmap, hashcat

- Wrote cross platform **reverse shell** and **team server** using Golang and WebSockets.
- Used **Donut** to generate **shellcode** which is received and run over HTTP on my own **loader**.
- Some experience with sandboxing, **static and dynamic code analysis**, port and vulnerability scanning, hash cracking, brute forcing, **reverse engineering**, and debugging web applications.

Web Development

Minecraft Protocol, Playwright, Chrome Extensions

- Built and maintained a **Discord bot** that delivered **real-time login notifications** of thousands of Hypixel players.
- Built a tool to **snipe Minecraft usernames** as soon as they become available.
- Wrote a **CAPTCHA** verification Discord bot for my personal Discord server.
- Collaborated with a team of five to build a **Chrome extension** using React.

Programming Languages

- C++, C#, Rust, Python, Java, Golang, TypeScript